# LINUX ADMINISTRATION

## FINAL PROJECT

### PHASE I: PROJECT DISCOVERY AND PROPOSAL

Real-world Linux systems investigation, expert interview,
and selection of practical project ideas

| ASSIGNMENT INFORMATION | DETAILS |
|---|---|
| Purpose | Help students choose a suitable Linux project for Phase II |
| Who to consult | An IT professional in an organization or a qualified individual freelancer |
| Project proposals | At least THREE suitable Linux project options |
| Phase I assessment | OPTIONAL and NOT GRADED |
| Preferred submission | ONE PDF through the lecturer's Google Form |
| Deadline | Sunday, 8 November 2026, at 5:00 AM (CAT, Rwanda time) |
| Phase II | One lecturer-approved Linux project, graded as the Final Project |

### Student guide and proposal requirements

Read the full guidance before choosing your projects.

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# Quick Guide: What You Need to Know

**IMPORTANT CHANGE: PHASE I IS OPTIONAL**

Phase I is provided to help you choose a practical Linux project that matches your interests and IT career path. It carries no marks. If you do not submit a proposal, the lecturer will assign a Phase II project to you. The approved Phase II project is the graded Final Project.

| QUESTION | ANSWER |
|---|---|
| Can I interview a freelancer? | Yes. The person must use Linux in real IT work, understand the project, and be able to explain and guide you on the project idea. |
| Must the expert work for a company? | No. A qualified individual freelancer, independent engineer, or consultant is acceptable. |
| How many projects should I propose? | At least three different project options. Every option must meet the minimum requirements in Section 7. |
| Who chooses my actual project? | The lecturer. One of your three proposals may be approved, refined, or replaced with a suitable project. |
| What if I do not submit Phase I? | You receive no Phase I mark or penalty. The lecturer assigns your Phase II project. |
| Do I implement anything in Phase I? | No. This phase is for investigation, understanding, and proposing realistic projects. |
| What is graded? | Phase II only, under the rubric and instructions issued for that phase. |

**YOUR ROUTE TO THE FINAL PROJECT**

1. Choose a real source: organization, institution, or qualified freelancer.
2. Interview the expert and identify Linux systems used in genuine IT work.
3. Select at least three project candidates that meet the minimum technical baseline.
4. Submit an optional project proposal as one PDF by the published deadline.
5. Receive a project assignment or approval from the lecturer.
6. Build, secure, test, document, and demonstrate the project in graded Phase II.

## 1. Purpose of the Assignment

Linux is used in software development, system administration, cybersecurity, networking, cloud services, databases, and many other areas of IT. This activity helps you discover how Linux solves real problems and choose a project that you can build with confidence.

You will speak with an experienced person who uses Linux, understand one or more real systems, and develop three possible projects. The lecturer will use your ideas to select a suitable Phase II project. A practical, well-explained idea is more useful than a very large project that cannot be completed.

### 1.1 Learning goals

- Connect Linux knowledge with your department, specialization, or IT career interests.
- Learn directly from engineers, developers, administrators, consultants, or freelancers.
- Explain what a Linux system does, how it works, and why it is used.
- Identify the Linux configuration, security, troubleshooting, and administration skills needed.
- Compare feasible project ideas and recommend projects suited to the available resources.
- Prepare a clear proposal that the lecturer can use to choose an appropriate final project.

### 1.2 What Phase I does not require

You are not required to build the project, obtain production access, carry out a security test, purchase software, or demonstrate a working system during Phase I. Formal practical implementation starts in Phase II after lecturer approval.

## 2. Who You May Interview

Students may investigate a Linux system through either of the following sources. Both are equally acceptable when the information is genuine and the selected projects meet the technical baseline.

### Option A. Organization or institution

Approach a company, bank, Internet service provider, school, university, hospital, government agency, IT department, data center, nonprofit organization, or similar workplace. Interview a professional involved in its Linux systems or applications.

### Option B. Individual freelancer or independent IT expert

You may interview an individual who independently uses Linux in daily professional IT work. Examples include freelance software developers, infrastructure consultants, DevOps specialists, cloud engineers, cybersecurity practitioners, database administrators, and network engineers.

#### CONDITIONS FOR AN ACCEPTABLE FREELANCER INTERVIEW

- The person uses Linux for real IT assignments, client work, personal professional services, or ongoing technical projects, not only for classroom exercises.
- The person can describe at least one real Linux-based system or project with enough technical detail to support an academic proposal.
- The person can explain the purpose, main components, workflow, Linux administration tasks, and expected results.

**FINDING THREE IDEAS**

You may obtain three suitable projects from one expert, or combine ideas from more than one organization or freelancer. For each idea, record its source. Do not invent systems or interview responses.

### 2.1 Meeting format and good practice

An interview can be conducted in person, by video call, or by telephone. Ask for consent, prepare your questions, take notes, and confirm key technical details. Where information is private, describe the system in general terms instead of asking for credentials or sensitive configurations.

## 3. How to Conduct Your Investigation

| STEP | WHAT TO DO | EXPECTED OUTPUT |
|---|---|---|
| 1. Find a source | Select an organization or qualified freelancer who genuinely uses Linux. | Contact details retained privately and a suitable interview arranged. |
| 2. Prepare | Review the expert's area of work and prepare questions. | Interview topics and a note-taking method. |
| 3. Interview | Ask about real Linux projects, their purpose, architecture, operations, and challenges. | Accurate interview notes and system descriptions. |
| 4. List systems | Record Linux services and projects that the expert can discuss. | An inventory of relevant Linux systems. |
| 5. Screen ideas | Use the mandatory minimum baseline in Section 7. | At least three acceptable project candidates. |
| 6. Compare | Rank your three choices by career relevance, learning value, and feasibility. | Three project proposals with clear justifications. |
| 7. Submit, if you choose | Prepare one PDF and upload it using the lecturer's Google Form. | A Phase I proposal for lecturer review, not grading. |

### 3.1 Core interview questions

#### PROFESSIONAL BACKGROUND

- What is your role, and how do you use Linux in daily work?
- Which Linux distributions, tools, platforms, or services do you use?
- Which three or more projects would you recommend to a Linux student?

#### SYSTEM AND BUSINESS PURPOSE

- What problem does each project solve, and who uses it?
- Why did you choose Linux or this solution?
- Which applications and components make the system work?
- How does information or network traffic move between the components?

#### LINUX ADMINISTRATION

- How is the system installed and configured?
- What commands or configuration files do you work with?
- How do you manage accounts, permissions, services, packages, and access?
- How do you monitor logs, performance, errors, and availability?
- How are backups, updates, and security controls handled?

**PLANNING A STUDENT PROJECT**

- Could the idea be implemented in an isolated lab or Linux virtual machine?
- What are the minimum resources and main configuration tasks?
- Which tests would show that the system is working and properly secured?
- What common errors should a student expect and learn to resolve?
- Could you provide general guidance if the student needs clarification later?

Use these as a guide, not a script. You may ask additional questions that fit your source and proposed area of specialization.

## 4. Linux Systems and Project Areas to Explore

You are free to propose projects from any suitable Linux field. These examples are intended to help you discover options, not to force every student into the same project. Confirm that Linux plays a real role in each idea.

| AREA / IT CAREER PATH | POSSIBLE LINUX SYSTEMS OR PROJECTS |
|---|---|
| System administration | Ubuntu Server, Debian, RHEL, Rocky Linux, user and group management, SSH, Linux hardening, systemd, KVM, Proxmox VE, Bash administration. |
| Network administration | BIND or Unbound DNS, Kea DHCP, WireGuard, OpenVPN, FreeRADIUS, Linux routing, nftables, Squid, HAProxy. |
| Cybersecurity | Wazuh, Security Onion, Suricata, Snort, Zeek, auditd, Fail2ban, vulnerability management, Linux SIEM, centralized logs. |
| Software development | NGINX or Apache HTTP Server, Python/Flask/Django, Node.js, PHP, Java/Tomcat, APIs, GitLab, Gitea, deployment servers. |
| DevOps and cloud | Docker or Podman on Linux, Kubernetes nodes, Jenkins, GitLab CI/CD, Ansible, Terraform, Linux cloud VMs, automated deployment. |
| Databases | PostgreSQL, MariaDB/MySQL, MongoDB, Redis, database roles, backups, replication, monitoring, Linux-hosted data services. |
| Monitoring and NOC | Zabbix, Nagios, LibreNMS, Prometheus, Grafana, rsyslog, Graylog, OpenSearch, service health checks and alerting. |
| Backup and storage | NFS, Samba, rsync, Restic, BorgBackup, Bacula, MinIO, Ceph, scheduled backups, backup verification and recovery. |
| Identity and access | FreeIPA, OpenLDAP, Samba AD, Keycloak on Linux, FreeRADIUS, SSH key-based access, centralized authentication. |
| Enterprise applications | Moodle, Odoo, ERPNext, Nextcloud, Postfix/Dovecot, internal web applications, institutional portals on Linux. |
| Research and embedded IT | Raspberry Pi Linux, edge gateways, IoT data services, Linux research computing, MQTT brokers, industrial data collection. |
| Automation and platform tools | Bash or Python automation, cron and systemd timers, Ansible playbooks, infrastructure configuration and recovery scripts. |

**CHECK THE OPERATING SYSTEM**

An open-source product is not automatically Linux-based. For example, pfSense and OPNsense are based on FreeBSD, not Linux. Projects must have Linux as a meaningful part of the implementation.

# 5. What to Document About Each Real System

For each system discussed during the interview, record what is known and may be shared. Keep descriptions short for minor tools and detailed for systems relevant to your project proposals.

| TOPIC | GUIDING QUESTIONS |
|---|---|
| Name and use | What is the project or service called? Who uses it and for what purpose? |
| Problem solved | What business, technical, or security problem does it address? |
| Linux environment | Which distribution or Linux platform does it use? Is it on a VM, physical server, cloud instance, or Linux-based device? |
| Main components | Which applications, services, users, clients, databases, and network connections are involved? |
| How it works | What happens from the first request or event to the final result? |
| Administration | Which installation, configuration, accounts, permissions, updates, service and troubleshooting tasks are required? |
| Security and operations | What general controls protect access? How are logs, monitoring, backups, and errors handled? |
| Benefits and constraints | Why is the solution useful? What are the costs, limitations, risks, or common challenges? |
| Student feasibility | How could a safe, smaller version be built and tested in a lab? |

## 5.1 Example of a meaningful description

Weak: "The developer uses NGINX and Linux to run a website."

Better: "The developer hosts a web application on Ubuntu Server. NGINX receives browser requests and forwards them to an application service. The application stores data in PostgreSQL. The administrator updates packages, configures user access, checks service logs, and backs up the database. The system supports an online client service and can be reproduced using Linux virtual machines for student practice."

The second example explains purpose, components, data flow, Linux tasks, and possible lab work. Use this level of thinking for the projects you want the lecturer to consider.

# 6. Selecting Three Project Candidates

Every student who chooses to submit Phase I must propose at least THREE distinct Linux project options that meet the baseline in Section 7. These are alternatives, not three projects that you must implement. The lecturer will choose ONE for graded Phase II.

## 6.1 Details required for each candidate

- A clear and specific project title, such as "Linux DNS and DHCP Services for a Small Office".

- The source of the idea: organization, institution, consultant, or freelancer, with a safe, non-sensitive description.
- A real problem, practical use case, or target users.
- Why the project matches your academic department, specialization, or preferred IT career.
- Linux distribution, services, tools, and how the components will connect.
- Four or more meaningful Linux administration competencies to be demonstrated.
- Security controls, operational management tasks, and troubleshooting expectations.
- A basic lab plan, required equipment or virtual machines, and available free or open-source software.
- At least four tests that will prove the main functions, security, and operations work.
- Expected practical deliverables and evidence for Phase II.
- Likely problems and a realistic plan to handle them.

### 6.2 Rank the candidates

Label the projects Choice 1, Choice 2, and Choice 3. Briefly explain which project you prefer and why. Your preferred choice is not automatically approved. The lecturer may select another option, adjust the scope, or assign a suitable alternative.

## 7. Minimum Requirements for an Acceptable Linux Final Project

The following baseline is the minimum technical standard for each of your three proposed projects. It is designed to be fair to students from different IT departments. A project does not need to be large, expensive, or complex, but it must give the student real experience in using and administering Linux.

**ALL PROJECT CANDIDATES MUST PASS THIS BASELINE**

A project that does not meet the minimum requirements will need to be improved or replaced before the lecturer can approve it for Phase II.

| MINIMUM REQUIREMENT | WHAT THE STUDENT MUST BE ABLE TO DO |
|---|---|
| 1. Real use and relevance | Solve a clear real-world IT problem and connect the solution to a field such as systems, networks, software, cloud, cybersecurity, databases, or IT operations. |
| 2. Linux is central | Use at least one Linux installation, VM, cloud instance, or Linux-based device as the main working environment. Perform meaningful tasks through the Linux command line. A container-only demonstration without Linux administration is insufficient. |
| 3. Working service and integration | Install and configure at least one useful Linux-hosted service or application, plus an interacting component such as a client, agent, database, second service, remote host, or automated workflow. |
| 4. At least four Linux skills | Demonstrate four meaningful Linux administration skills. They must include service setup/configuration, user access or system security, logging/troubleshooting, and at least one skill related to the chosen specialization. |

| MINIMUM REQUIREMENT | WHAT THE STUDENT MUST BE ABLE TO DO |
|---|---|
| 5. Basic security | Apply and explain at least one appropriate security control, such as least-privilege permissions, SSH key use, firewall rules, TLS, secure authentication, or account restrictions. |
| 6. Operational management | Include at least one ongoing administration activity, such as health monitoring, log review, backup and restore, scripting, scheduled maintenance, service recovery, or configuration automation. |
| 7. Testable results | Define at least four practical tests, including a main function test, an access/security test, an operational or troubleshooting test, and an integration or recovery test. The tests must have clear expected results. |
| 8. Feasible and safe | Use a legal, authorized student lab. Prefer free tools and manageable resources. Avoid dependence on confidential information, company production access, expensive licenses, or an external expert doing the implementation. |
| 9. Demonstrable evidence | Be able to show configuration steps or files, relevant commands, test results, screenshots or logs with private data hidden, an architecture diagram, and a short explanation of how the system works. |

## 7.1 Four required skills, explained

The four skills should be visible in the proposed design, not mentioned only as learning goals:

- Service setup and configuration: install a Linux application, configure its service files or packages, and start or enable it correctly.
- Access or security: create suitable accounts, manage file permissions, configure authentication, apply firewall rules, or use another relevant control.
- Logging and troubleshooting: inspect logs, check service status, diagnose at least one error, and explain how to restore normal operation.
- Specialization skill: choose a task connected to your field, such as DNS, VPN, deployment, containers, SIEM alerts, database backups, API hosting, monitoring, or automation.

A strong project may demonstrate many more skills. The lecturer may accept an equivalent design where these skills are shown through different but comparable technical activities.

## 7.2 Projects that are too limited

- Installing Ubuntu or another Linux distribution without configuring useful services.
- Running a few basic commands and presenting screenshots without a working solution.
- Writing only a literature review, a system description, or a presentation.
- Deploying a website only through a hosting dashboard with no Linux administration.
- Using a non-Linux product as the entire project without a meaningful Linux component.
- Copying configuration files or scripts without understanding, testing, and explaining them.

## 8. Example Project Options by Career Path

These are examples of acceptable starting points when their scope meets Section 7. You may adapt them or propose different projects based on your interviews and professional interests.

| SPECIALIZATION | EXAMPLE PROJECT | PRACTICAL LINUX SKILLS AND TESTS |
|---|---|---|
| System administration | Secure Linux file sharing for a small office | Samba or NFS, groups and permissions, service administration, logs, backup and client access tests. |
| Computer networks | Linux DNS and DHCP server for a lab network | BIND or Unbound with Kea DHCP, interface configuration, firewall, client name-resolution and lease tests. |
| Cybersecurity | Centralized security event monitoring | Wazuh manager or comparable service, Linux agent, safe test events, alert review, logs, hardening. |
| Software engineering | Linux web application with a database | NGINX or Apache, Python/Node.js/PHP, database connection, account permissions, logs, TLS or firewall. |
| DevOps | Automated deployment of a Linux-hosted app | Linux host, application container or service, CI pipeline or Ansible, secrets handling, rollbacks or monitoring. |
| Database administration | Secure PostgreSQL database service | Linux PostgreSQL server, roles, access restrictions, remote or local client, scheduled backup and restore tests. |
| NOC / operations | Centralized monitoring for two Linux hosts | Zabbix or Prometheus, monitored endpoints, service checks, alerts, logs, dashboard and failure test. |
| Cloud administration | Secure cloud Linux service with automation | Cloud Linux VM, SSH access controls, deployed service, startup automation, monitoring and recovery tests. |
| Embedded / IoT | Linux sensor data collection and dashboard | Linux device or VM, MQTT or API service, client/sensor simulator, logs, access control and data flow tests. |

Example technologies do not have to be used exactly as listed. Select tools suited to your resources and learning goals. Successful configuration, testing, and explanation matter more than the size of the toolset.

## 9. Optional Phase I Report: Recommended Format

Students who would like the lecturer to consider their preferred project ideas should submit one clear PDF. A report of about 6 to 12 pages, excluding optional appendices, is normally sufficient when it contains useful technical detail.

| SECTION | WHAT TO INCLUDE |
|---|---|
| Cover page | Institution, Linux Administration, Phase I, full name, registration number, program/class, date. |
| 1. Short introduction | What you investigated, your preferred career area, and why you selected the interview source. |

| SECTION | WHAT TO INCLUDE |
|---|---|
| 2. Interview source | Organization or freelancer profile, general expertise, interview date and method. Use anonymity where required. |
| 3. Linux systems identified | A brief table of real systems or projects discussed with the expert. Include role and main Linux technologies. |
| 4. Three project proposals | Choice 1, Choice 2, Choice 3. For EACH: problem, objectives, Linux architecture, services, skills, security, tests, resources, expected result, and feasibility. |
| 5. Project comparison | Explain how all three satisfy the Section 7 baseline and compare effort, career relevance, and available resources. |
| 6. Preferred option | Recommend one candidate and give a short technical justification. |
| 7. References and notes | Cite public documentation and add any approved diagrams or anonymous interview notes. |

### 9.1 Writing and presentation

- Use simple academic English, short paragraphs, clear headings, and accurate technical terms.
- Describe what you actually learned from the interview; do not paste generic vendor descriptions.
- Use meaningful tables or simple diagrams where they help explain system components.
- Give credit to websites, manuals, and other published materials you consulted.
- Use 11 or 12 point body text, consistent formatting, and page numbers.
- Export a single searchable PDF and check every page before uploading it.

## 10. Submission and Lecturer Project Selection

| ITEM | RULE |
|---|---|
| Phase I submission | Optional. Students may submit an investigation and three project ideas for guidance. |
| Phase I grade | No marks. The lecturer will review submissions for suitability, not grade them. |
| Submission channel | Google Form provided by the lecturer. |
| Deadline | Sunday, 8 November 2026, at 5:00 AM (CAT, Rwanda time, UTC+2). |
| Upload format | One PDF file containing the entire proposal and any permitted appendices. |
| Suggested file name | REGISTRATION_NUMBER_LINUX_PHASE1_PROJECT_PROPOSAL.pdf |
| When submitted | The lecturer reviews the three candidates and selects one, modifies its scope, or assigns a more suitable project. |
| When not submitted | The lecturer assigns a Phase II Linux project. No Phase I grading penalty applies. |
| Phase II assessment | The approved or assigned implementation is graded as the Final Project. The Phase II requirements and marking criteria will be issued separately. |

**ONE PROJECT WILL BE GRADED**

Phase I supports project choice. Phase II is the actual Final Project assessment. Every student must complete one lecturer-approved or lecturer-assigned Linux project in Phase II.

### 10.1 Lecturer acceptance checklist

The lecturer will use professional judgment rather than Phase I marks. A project is more likely to be approved when it:

- Is based on a real IT need or useful professional scenario.
- Uses Linux as a central platform and requires command-line administration.
- Meets every baseline criterion in Section 7 or an approved equivalent.
- Fits the student's department or desired IT career while still developing Linux skills.
- Can be completed and demonstrated with reasonable time, computing resources, and safe access.
- Has clear technical objectives, expected results, and tests.

## 11. Ethics, Privacy, and Academic Integrity

Respect the time and confidentiality of the people who help you. The purpose of this activity is learning, not accessing or testing anyone's production environment.

- Do not ask for passwords, private keys, customer data, proprietary code, or confidential configurations.
- Do not scan, attack, change, or log into production systems without separate explicit authorization. No such access is needed for Phase I.
- Get permission before recording a meeting, taking photographs, publishing names, or sharing screenshots.
- A freelancer may describe a client system without identifying the client or exposing private information.
- Use anonymized examples and simplified diagrams when an expert cannot share internal details.
- Do not invent interviews, project use cases, or technologies. Clearly identify what came from an interview and what came from public research.
- The interview source may offer advice, but the student must carry out and explain the Phase II project personally.

## 12. Phase II: What Comes Next

After reviewing Phase I reports, the lecturer will confirm one final project for each student. Students who did not submit Phase I will receive an assigned project. All students will be assessed on the implementation work under the Phase II instructions, not on whether they completed the optional Phase I proposal.

The Phase II instructions will define the practical scope, deadline, evidence, presentation expectations, and grading rubric. In general, the final project will require students to prepare a Linux environment, configure the approved solution, apply basic security, test it, solve problems, and explain their work.

**FINAL GUIDANCE**

Choose a project that is useful in your field and realistic to finish. A small Linux solution that is correctly installed, secured, tested, and explained is better than a large unfinished project.

## Appendix A. Interview Notes Template

| FIELD | STUDENT NOTES |
|---|---|
| Source type | Organization / institution / freelancer / independent consultant |
| Source or expert reference | Name or approved anonymous identifier |
| Professional role | Job title, specialization, or freelance area |
| Interview date and method | Date, in person / video / phone |
| Linux systems discussed | List key technologies or projects |
| Main problems solved | Summarize in your own words |
| Most useful insights | Linux skills, tools, security, or operational tasks |
| Recommended student projects | List ideas and key requirements |
| Guidance available | General advice offered by the source, if any |
| Privacy restrictions | What information cannot be disclosed |

## Appendix B. Linux System Investigation Sheet

Complete one sheet for each significant real Linux system you investigate.

| FIELD | YOUR DESCRIPTION |
|---|---|
| System or project name | |
| Source and professional area | |
| Purpose and users | |
| Problem solved and benefits | |
| Linux distribution or environment | |
| Main applications and components | |
| Basic architecture and workflow | |
| Linux installation and administration tasks | |
| Security and access controls | |
| Logging, monitoring, backup, or recovery | |
| Common technical challenges | |
| Possible student lab version | |

"Dream big and work hard to achieve your goals. Stay strong and keep fighting for your dreams. Do not let anything discourage you. The future looks bright."

# Good luck!